

Here is a readme for the data in the `Oil_capacity.xlsx` file:

Tab: Tanker

- **ship type:** The category or classification of the tanker.
- **min length (m):** The minimum length of the ship type in meters.
- **max length (m):** The maximum length of the ship type in meters.
- **Silo ship size (m):** The specific size of the silo ship in meters.
- **cargo capacity SILO (liters):** The cargo capacity of the SILO ship in liters.
- **cargo capacity (liters):** The total cargo capacity of the ship in liters.
- **hfo fuel capacity (liters):** The capacity for Heavy Fuel Oil (HFO) in liters.
- **diesel fuel capacity (liters):** The capacity for diesel fuel in liters.
- **total fuel capacity (liters):** The combined total fuel capacity in liters.
- **Data Source (all but "f"):** The data source for all columns except column "f".
- **Data Source Column "f":** The data source specifically for column "f".

Tab: atb

- **MMSI:** This column contains the Maritime Mobile Service Identity (MMSI) for each vessel, which is a series of nine digits used to uniquely identify ships.
- **TUG_NAME:** This column lists the name of the tugboat.
- **AIS ship tracks:** This column likely indicates the number of AIS (Automatic Identification System) ship tracks recorded for the vessel.
- **Median ship track distance (km):** This column represents the median distance of the AIS ship tracks in kilometers.
- **Include?:** This column appears to be a flag indicating whether to include AIS ship tracks, with 1 likely meaning 'yes' and 0 meaning 'no'.
- **AIS ship tracks to include:** This column seems to specify the number of AIS ship tracks that are to be included in the analysis.
- **ECY Oil transfers (2018):** This column records the number of Department of Energy (DOE) oil transfers that occurred in 2018.
- **Married tank barge name:** This column provides the name of the tank barge that is "married" (i.e., partnered) with the tugboat.
- **Vessel type:** This column describes the type of vessel.
- **Vessel length (m):** This column contains the length of the vessel reported online, in meters.
- **AIS vessel length (m):** This column contains the vessel length as reported by the AIS system, in meters.
- **AIS vessel length correction (m):** Corrected vessel length, in meters, to make length metric the same across vessels, changing those reporting only tug length to have lengths inclusive of barge.
- **Web/AIS confirmed barge/tug "marriage":** This column indicates whether the partnership between the tug and tank barge has been confirmed through web sources or AIS data.
- **HFO Capacity (liters):** This column lists the capacity for Heavy Fuel Oil (HFO) in liters.
- **Diesel Capacity (liters):** This column lists the capacity for Diesel fuel in liters.

- **Total Fuel Capacity (liters):** This column represents the total fuel capacity (HFO + Diesel) in liters.
- **Cargo Capacity (liters):** This column contains the cargo capacity of the tank barge in liters.
- **Number of cargo holds:** This column indicates the number of cargo holds on the tank barge.
- **Capacity Notes:** This column includes additional notes regarding the capacity of the vessel, such as "at 96% full".
- **Tug Source:** This column specifies the source of the information used for the tug and tank barge data.

Tab: cargo

- **ship type:** Specifies the category or type of the vessel (e.g., Bulk carrier, Panamax container).
- **vessel length (m):** Indicates the length of the vessel in meters.
- **fuel capacity (gallons):** Shows the fuel storage capacity of the vessel in US gallons.
- **fuel capacity (liters):** Shows the fuel storage capacity of the vessel in liters.
- **data source:** Identifies the source from which the data for the vessel information was obtained.

Tab: cruise

- **ship type:** Specifies the category or type of the cruise vessel (e.g., Norwegian Spirit, Freedom of the Seas).
- **vessel length (m):** Indicates the length of the cruise vessel in meters.
- **fuel capacity (gallons):** Shows the fuel storage capacity of the cruise vessel in US gallons.
- **fuel capacity (liters):** Shows the fuel storage capacity of the cruise vessel in liters.
- **data source:** Identifies the source from which the data for the cruise vessel information was obtained.

Tab: ferry

- **WA state ferry vessel type:** Specifies the category or type of the ferry vessel (e.g., Jumbo Class, Super Class).
- **vessel length (m):** Indicates the length of the ferry vessel in meters.
- **fuel capacity (gallons):** Shows the fuel storage capacity of the ferry vessel in US gallons.
- **fuel capacity (liters):** Shows the fuel storage capacity of the ferry vessel in liters.
- **data source:** Identifies the source from which the data for the ferry vessel information was obtained.

Tab: fishing

- **vessel length (m):** Indicates the length of the fishing vessel in meters.
- **fuel capacity (gallons):** Shows the fuel storage capacity of the fishing vessel in US gallons.
- **fuel capacity (liters):** Shows the fuel storage capacity of the fishing vessel in liters.
- **data source:** Identifies the source from which the data for the fishing vessel information was obtained.

Tab: smallpass data

- **MMSI:** The Maritime Mobile Service Identity number, a unique identifier for the vessel.
- **COUNT:** The number of occurrences or records associated with the MMSI.
- **LENGTH:** The length of the vessel.
- **Classified as "Small Passenger" in Krista's List:** A binary indicator (1 or 0) showing if the vessel is classified as a "Small Passenger" in Krista's List.
- **AIS min length:** The minimum length reported for the vessel according to AIS (Automatic Identification System) data.
- **AIS max length:** The maximum length reported for the vessel according to AIS data.
- **AIS median length:** The median length reported for the vessel according to AIS data.
- **AIS mean length:** The mean (average) length reported for the vessel according to AIS data.
- **total AIS tracks:** The total number of AIS tracks recorded for the vessel.
- **Total number of vessels:** The total count of vessels.

Tab: * data

The remaining tabs with “data” in title specify the MMSI grouped into each category. An example of the columns in these tabs is:

- **MMSI:** This column contains the Maritime Mobile Service Identity (MMSI), a unique identifier for the vessel.
- **COUNT:** This column represents the number of AIS pings or data points recorded for a given MMSI in 2018.
- **LENGTH:** This column indicates the length of the vessel, in meters.
- **Median time between pings (minutes):** This column shows the median time interval, in minutes, between consecutive AIS pings for the vessel.
- **Standard deviation (minutes):** This column provides the standard deviation of the time between AIS pings, in minutes, indicating the variability of ping intervals.
- **total AIS tracks:** This column shows the total number of AIS tracks recorded for the vessel.
- **Total number of vessels:** This column indicates the total number of vessels represented in the dataset.

Tab: vessel classification

- **MIDOSS Category:** This column categorizes vessel groupings used in the MIDOSS analysis (e.g., Tankers, ATB, cargo, cruise, ferry, fishing, small passenger).
- **Common name(s):** This column lists common names or specific types of vessels within each MIDOSS category (e.g., tanker, bulk carrier, WA state ferries, tour boat).
- **AIS ship type:** This column specifies the general AIS ship type classification for the vessel (e.g., tanker, tug, cargo, Passenger, fishing).
- **AIS coarse type:** This column provides the AIS coarse type number or range, which is a more granular classification of the vessel type according to AIS standards.
- **Min observed length:** This column indicates the minimum observed length of vessels in that category. The unit is not specified but is typically meters.
- **Max observed length:** This column indicates the maximum observed length of vessels in that category. The unit is not specified but is typically meters.
- **Capacity Reference:** This column refers to information sources listed in tab “data source”.

- **Notes:** This column provides additional explanatory notes or details regarding the vessel category, its classification, or data attribution.